

Logic in Computer Science
Problem Sheet 1: Propositional Logic Syntax
August 23, 2025

1. A *statement* is a declarative sentence, conveying a definite meaning which may be either true or false but not both simultaneously. Incomplete sentences, questions and exclamations are not statements. Which of the following are statements? Why?
 - (a) If it rains, the streets get wet.
 - (b) If it rains, the streets remain dry.
 - (c) It rains.
 - (d) If it rains.
 - (e) John is intelligent and John is not intelligent.
 - (f) If John is intelligent then John is intelligent.
 - (g) If John is intelligent then John is not intelligent.
 - (h) This sentence is false.
 - (i) This sentence is true.
 - (j) There is life outside our solar system.
2. Write derivation sequences for the following well formed formulas (*wff*s).
 - (a) $((p_1 \rightarrow p_2) \leftrightarrow (p_1 \rightarrow p_3)) \rightarrow p_3$
 - (b) $((p_1 \wedge p_2) \rightarrow p_3) \leftrightarrow ((p_1 \rightarrow p_2) \vee (p_1 \rightarrow p_3))$
 - (c) $((p_1 \rightarrow p_3) \rightarrow (p_2 \rightarrow p_3)) \rightarrow ((p_1 \vee p_2) \rightarrow p_3)$
 - (d) $(\neg(p_1 \vee ((\neg p_1) \wedge p_2)))$
3. Write formation trees for the *wff*s in question 3.
4. Write the set of all subformulas of the *wff*s in question 3.
5. Write the *wff*s in question 3 using prefix and postfix notation.
6. Write three different derivation sequences for $(p \rightarrow q)$.
7.
 - (a) Define recursively the set of all subformulas of A which have negation as the principal operator.
 - (b) Define recursively the set of all atomic propositions occurring in A .
 - (c) Define recursively the prefix notation of a formula A .
 - (d) Define recursively the number of boolean operators occurring in a formula A .
 - (e) Define recursively the nested depth of the formula A (maximum number of internal nodes in any path from root node to leaf node of the formation tree of A).
8. Are there propositions with length 1, 2, ...?